

DES MOINES 17 E, IA, USA

WMO: 744480

Lat: 41.5562N

Lon: 93.2855W

Elev: 281

StdP: 98.00

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54902

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.0	-18.4	-24.4	0.4	-20.7	-22.0	0.5	-17.9	10.7	-7.2	9.3	-6.0	2.5	N/A	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.6	24.5	31.2	24.1	29.7	23.3	26.2	30.6	25.1	29.3	24.1	28.5	2.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.7	29.6	23.9	19.4	28.1	22.8	18.1	27.0	83.1	30.7	78.2	29.4	74.1	28.5	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.3	6.3	-25.5	34.8	3.0	2.0	-27.7	36.2	-29.5	37.4	-31.2	38.5	-33.4	40.0
			-25.9	27.8	3.2	0.7	-28.2	28.3	-30.1	28.7	-31.8	29.1	-34.1	29.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-5.6	-4.3	3.7	10.4	16.3	21.6	23.4	22.1	18.6	11.2	4.0	-2.9
	DBStd	11.56	7.15	7.03	6.68	5.48	4.62	3.01	3.15	2.85	4.38	5.22	5.86	6.38
	HDD10.0	1810	485	401	217	60	6	0	0	0	1	48	192	400
	HDD18.3	3543	743	633	454	241	94	7	2	3	50	227	429	658
	CDD10.0	1793	0	1	22	73	201	347	416	376	258	86	13	1
	CDD18.3	485	0	0	1	4	31	104	159	121	57	8	0	0
	CDH23.3	4147	0	0	11	63	251	851	1461	963	479	66	1	0
	CDH26.7	1307	0	0	1	12	58	251	534	291	147	12	0	0
(14) Wind	WSAvg	2.7	3.3	3.2	3.3	3.6	2.9	2.2	1.8	1.7	2.0	2.6	2.8	2.9
(15) Precipitation	PrecAvg	914	24	28	53	96	127	143	121	109	92	58	48	27
	PrecMax	1308	50	53	149	202	214	312	298	302	174	160	134	73
	PrecMin	692	10	5	1	36	25	33	22	13	40	7	1	2
	PrecStd	167	9	15	38	41	45	70	73	70	37	41	34	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	17.5	25.0	28.6	31.4	32.7	35.7	33.9	32.4	28.6	21.9	14.4
		MCWB	6.8	10.8	17.3	18.8	22.8	24.7	25.0	23.3	23.0	19.6	14.3	9.7
	2%	DB	8.7	12.2	20.9	25.0	28.1	31.2	33.2	31.4	30.4	25.1	18.4	10.6
		MCWB	5.7	7.3	13.9	16.3	20.5	24.1	25.3	23.8	23.1	17.4	12.8	7.7
	5%	DB	5.9	8.3	17.4	22.4	26.0	29.6	31.5	29.7	28.3	22.6	15.4	8.4
		MCWB	3.3	5.2	11.9	14.7	18.9	23.1	25.0	23.3	21.6	16.4	10.5	5.9
	10%	DB	3.6	5.2	14.0	19.8	24.1	28.0	29.7	28.1	26.0	20.1	12.9	5.6
		MCWB	1.8	3.0	9.7	12.8	18.3	22.2	24.3	22.4	20.1	15.2	8.8	3.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	12.9	17.8	19.4	23.7	27.0	27.6	27.0	24.9	21.3	15.2	12.0
		MCDB	11.7	16.5	24.2	26.2	30.1	31.6	32.6	31.3	30.5	24.2	19.0	13.2
	2%	WB	5.2	8.5	15.4	17.3	22.0	25.4	26.4	25.2	23.9	19.4	13.3	8.9
		MCDB	8.3	11.9	19.2	23.4	27.4	29.9	31.1	28.8	29.0	22.2	17.4	9.9
	5%	WB	3.4	5.5	12.7	15.6	20.7	24.1	25.5	24.0	22.9	17.7	11.0	5.9
		MCDB	5.5	8.6	16.6	21.1	25.1	28.0	29.8	28.3	27.1	20.3	14.4	8.5
	10%	WB	1.8	3.1	9.9	13.9	19.5	23.0	24.5	23.1	21.7	15.4	9.0	3.6
		MCDB	3.5	5.4	14.0	18.3	23.2	27.0	28.7	27.1	24.9	19.4	12.3	5.7
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.6	10.6	11.7	10.6	10.3	10.6	10.6	11.7	11.1	10.2	8.7
		MCDBR	11.9	13.7	15.0	15.5	12.6	11.6	12.1	12.1	13.0	13.9	13.8	11.7
	5% WB	MCWBR	8.8	9.4	9.1	8.9	6.6	6.5	5.7	6.1	6.1	7.6	8.7	8.2
		MCDBR	11.3	13.9	13.4	13.8	11.5	10.7	10.2	10.7	10.8	11.0	12.9	10.5
		MCWBR	8.6	10.0	8.1	8.4	6.5	6.8	5.9	5.9	5.7	7.5	9.2	8.1
(40) Clear-Sky Solar Irradiance	taub	0.290	0.315	0.331	0.383	0.409	0.421	0.438	0.419	0.385	0.343	0.301	0.290	
	taud	2.416	2.346	2.396	2.271	2.266	2.279	2.219	2.287	2.335	2.448	2.498	2.432	
	Ebn at Noon	865	890	918	887	868	855	836	842	847	850	848	832	
	Edh at Noon	79	99	106	129	133	132	139	126	112	89	72	72	
(44) All-Sky Solar Radiation	RadAvg	1.92	2.92	3.79	4.74	5.59	6.21	6.27	5.44	4.59	3.07	2.09	1.60	
	RadStd	0.18	0.20	0.36	0.36	0.43	0.61	0.47	0.35	0.34	0.35	0.18	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page